

Lab Manual : HTML

LAB 1: Introduction to HTML & Basic Structure

1. Objective

- Understand basic HTML structure
- Learn how a webpage is created

2. Theory

HTML (HyperText Markup Language) is used to create webpages. Every HTML document follows a basic structure.

3. Main Tags

`<!DOCTYPE html>, <html>, <head>, <title>, <body>`

4. Syntax

Basic structure:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Page Title</title>
  </head>
  <body>
    All the tags related contents will be here
  </body>
</html>
```

5. FULL HTML CODE

```
<!DOCTYPE html> <!-- Defines HTML5 document -->
<html> <!-- Root element -->
  <head> <!-- Contains meta information -->
    <title>My First Web Page</title> <!-- Page title -->
  </head>

  <body> <!-- Visible content starts -->
    <h1>Welcome to HTML</h1>
    <p>This is my first web page.</p>
  </body>
</html>
```

Output :

Welcome to HTML

This is my first web page.

6. LINE-BY-LINE EXPLANATION

- *<!DOCTYPE html> → tells browser it's HTML5*
- *<html> → root of webpage*
- *<head> → contains title and metadata*
- *<title> → shown in browser tab*
- *<body> → visible part of webpage*
- *<h1> → heading*
- *<p> → paragraph*

7. Output Description

- A page with heading “Welcome to HTML”
- A paragraph below it

8. Practice Task

- Change title to your name

- Add another paragraph

9. Project - 1

Create a personal introduction page

LAB 2: Basic Tags (Heading, Paragraph, Line Break, HR)

1. Objective

Learn basic text tags

2. Theory

HTML provides tags to structure content like headings and paragraphs.

3. Tags Covered

<h1> to <h6>, <p>,
, <hr>

4. FULL HTML CODE

```
<!DOCTYPE html>
<html>
<head>
  <title>Basic Tags</title>
</head>
<body>
  <h1>Main Heading</h1>
  <h2>Sub Heading</h2>
  <p>This is a paragraph.</p>
  <p>This is another line.<br>Line break used here.</p>
  <hr>
  <p>Content after horizontal line.</p>
</body>
</html>
```

Output :

Main Heading

Sub Heading

This is a paragraph.

This is another line.
Line break used here.

Content after horizontal line.

5. LINE-BY-LINE EXPLANATION

- `<h1>` → largest heading
- `<h2>` → smaller heading
- `<p>` → paragraph
- `<br>` → breaks line
- `<hr>` → draws horizontal line

6. Output Description

- Headings displayed
- Paragraph with line break
- Horizontal line

7. Student Task

- Use all heading tags (h1–h6)
- Add 3 paragraphs with your name, address and college name

8. Real-World Project

Create school notice page with heading and paragraph

LAB 3: Formatting Tags

1. Objective

Learn how to style text

2. Theory

Formatting tags help emphasize text

3. Tags Covered

`<b>`, `<i>`, `<u>`, `<strong>`, `<em>`, `<mark>`, `<small>`, `<del>`, `<ins>`, `<sup>`,
`<sub>`

4. FULL HTML CODE

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Formatting Tags</title>
```

```
</head>
```

```
<body>
```

```
  <p><b>Bold Text</b></p>
```

```
  <p><i>Italic Text</i></p>
```

```
  <p><u>Underlined Text</u></p>
```

```
  <p><strong>Important Text</strong></p>
```

```
  <p><em>Emphasized Text</em></p>
```

```
  <p><mark>Highlighted Text</mark></p>
```

```
  <p><small>Small Text</small></p>
```

```
  <p><del>Deleted Text</del></p>
```

```
  <p><ins>Inserted Text</ins></p>
```

```
  <p>H<sub>2</sub>O</p>
```

```
  <p>X<sup>2</sup></p>
```

Bold Text

Italic Text

Underlined Text

Important Text

Emphasized Text

Highlighted Text

Small Text

~~Deleted Text~~

Inserted Text

H₂O

x²

</body>

</html>

5. LINE-BY-LINE EXPLANATION

- → bold
- <i> → italic
- <u> → underline
- → important (bold semantic)
- → emphasis
- <mark> → highlight
- <sub> → subscript
- <sup> → superscript

6. Output Description

Different styled texts displayed

7. Student Task

Write chemical formulas and math equations

8. Real-World Project

Create science notes webpage

LAB 4: List Tags

1. Objective

Learn how to create lists

2. Theory

Lists organize content

3. Tags Covered

`<ul>`, `<ol>`, `<li>`, `<dl>`, `<dt>`, `<dd>`

4. FULL HTML CODE

```
<!DOCTYPE html>
<html>
<head>
  <title>Lists</title>
</head>
<body>

  <h2>Unordered List</h2>
  <ul>
    <li>Apple</li>
    <li>Banana</li>
  </ul>

  <h2>Ordered List</h2>
  <ol>
    <li>Step 1</li>
    <li>Step 2</li>
```


<h2>Description List</h2>

<dl>

 <dt>HTML</dt>

 <dd>Markup language</dd>

</dl>

</body>

</html>

Output:

Unordered List

- Apple
- Banana

Ordered List

1. Step 1
2. Step 2

Description List

HTML
 Markup language

5. LINE-BY-LINE EXPLANATION

- → unordered list
- → ordered list
- → list item
- <dl> → description list

- `<dt>` → term
- `<dd>` → description

7. Student Task

- Create shopping list
- Create steps of cooking Chicken

8. Real-World Project

Create restaurant menu page

LAB 5: Image Tag

1. Objective

- Learn how to insert images in a webpage

2. Theory

The `<img>` tag is used to display images. It is a self-closing tag.

3. Tags Covered

`<img>`

Attributes: `src`, `alt`, `width`, `height`

4. Syntax Explanation

```

```

5. FULL HTML CODE

```
<!DOCTYPE html>
<html>
<head>
  <title>Image Example</title>
</head>
<body>

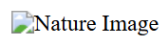
  <h2>My Image</h2>

  <!-- Display image -->
  

</body>
</html>
```

Output:

My Image



6. LINE-BY-LINE EXPLANATION

- `<img>` → inserts image
- `src` → image path
- `alt` → text if image not loaded
- `width/height` → size of image

7. Output Description

Image displayed with given size

8. Student Task

- Add 3 different images
- Change sizes

9. Real-World Project

Create school photo section

LAB 6: Image Gallery (Mini Project)

1. Objective

- Create a simple image gallery

2. Theory

Multiple images can be arranged to form a gallery.

3. Tags Covered

, <div>

4. FULL HTML CODE

```
<!DOCTYPE html>
<html>
<head>
  <title>Image Gallery</title>
</head>
<body>

  <h2>My Gallery</h2>
  <div>
    
    
    
  </div>
</body>
</html>
```

Output:

My Gallery



5. LINE-BY-LINE EXPLANATION

- `<div>` → container for images
- Multiple `<img>` → gallery

6. Output Description

Images displayed side by side

7. Student Task

- Add 5 images
- Add captions using `<p>`

8. Real-World Project

Create college event gallery

LAB 7: Marquee Tag

1. Objective

- Learn simple animation

2. Theory

`<marquee>` scrolls text (not recommended in modern HTML but used in syllabus)

3. Tags Covered

`<marquee>`

4. FULL HTML CODE

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Marquee</title>
```

```
</head>
```

```
<body>
```

```
  <marquee>Welcome to My Website</marquee>
```

```
</body>
```

```
</html>
```

Output:

Welcome to My Website

5. LINE-BY-LINE EXPLANATION

- `<marquee>` → scrolling text

6. Output Description

Text moves across screen

7. Student Task

- Change direction
- Add color text

8. Real-World Project

Create school announcement banner

LAB 8: Table Tag

1. Objective

- Learn to create tables

2. Theory

Tables organize data in rows and columns.

3. Tags Covered

`<table>`, `<tr>`, `<th>`, `<td>`

4. FULL HTML CODE

```
<!DOCTYPE html>
<html>
<head>
  <title>Table</title>
</head>
<body>
```

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Ram</td>
    <td>18</td>
  </tr>
</table>
```

```
</body>  
</html>
```

Output :

Name	Age
Ram	18

5. LINE-BY-LINE EXPLANATION

- `<table>` → table start
- `<tr>` → row
- `<th>` → heading
- `<td>` → data

6. Output Description

Simple table displayed

7. Student Task

- Create student table
- Add 5 rows

8. Real-World Project

Create result sheet

LAB 9: Advanced Table (Rowspan & Colspan)

1. Objective

- Learn table merging

2. Theory

Rowspan and colspan merge cells

3. Tags Covered

rowspan, colspan

4. FULL HTML CODE

```
<!DOCTYPE html>
<html>
<head>
  <title>Advanced Table</title>
</head>
<body>

<table border="1">
  <tr>
    <th colspan="2">Student Info</th>
  </tr>
  <tr>
    <td rowspan="2">Ram</td>
    <td>18</td>
  </tr>
  <tr>
    <td>Grade A</td>
  </tr>
</table>

</body>
</html>
```

Output :

Student Info	
Ram	18
	Grade A

5. LINE-BY-LINE EXPLANATION

- colspan="2" → merge columns
- rowspan="2" → merge row

6. Output Description

Merged table cells

7. Student Task

- Create timetable

8. Real-World Project

Create class routine table

LAB 10: Anchor Tag (Links)

1. Objective

- Learn linking between pages

2. Theory

Anchor tag connects webpages

3. Tags Covered

<a>

4. FULL HTML CODE

```
<!DOCTYPE html>
<html>
<head>
  <title>Links</title>
</head>
<body>
  <a href="https://www.google.com">Go to Google</a><br>

  <a href="about.html">About Page</a><br>
  <a href="mailto:example@gmail.com">Send Email</a>

</body>
</html>
```

Output :

[Go to Google](https://www.google.com)
[About Page](about.html)
[Send Email](mailto:example@gmail.com)

5. LINE-BY-LINE EXPLANATION

- <a> → link
- href → destination
- mailto: → email link

6. Output Description

Clickable links displayed

7. Student Task

- Link 2 pages
- Add email link

8. Real-World Project

Create school website navigation

LAB 11: Introduction to Forms

1. Objective

- Learn how to collect user input

2. Theory

Forms are used to collect data from users like name, email, etc.

3. Tags Covered

`<form>`, `<input>`, `<label>`

4. FULL HTML CODE

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Form</title>
</head>
<body>

<form>
  <label>Name:</label>
  <input type="text"><br><br>

  <label>Email:</label>
  <input type="email"><br><br>

  <input type="submit" value="Submit">
</form>
```

```
</body>  
</html>
```

Output :

Name:

Email:

5. LINE-BY-LINE EXPLANATION

- `<form>` → form container
- `<label>` → field name
- `<input>` → input field
- `type="text"` → text input
- `type="email"` → email input
- `submit` → button

6. Output Description

Simple form with input fields

7. Student Task

- Add phone number field
- Add password field

8. Real-World Project

Create feedback form

LAB 12: Form Elements

1. Objective

- Learn advanced form inputs

2. Tags Covered

radio, checkbox, select, textarea

3. FULL HTML CODE

```
<!DOCTYPE html>
<html>
<body>

<form>
  Gender:
  <input type="radio" name="g">Male
  <input type="radio" name="g">Female<br><br>

  Hobbies:
  <input type="checkbox">Reading
  <input type="checkbox">Sports<br><br>

  City:
  <select>
    <option>Kathmandu</option>
    <option>Pokhara</option>
  </select><br><br>

  Message:
  <textarea></textarea><br><br>

  <input type="submit">
</form>

</body>
</html>
```

Output :

Gender: ☐ Male ☐ Female

Hobbies: ☐ Reading ☐ Sports

City:

Message:

4. LINE-BY-LINE EXPLANATION

- radio → choose one
- checkbox → choose multiple
- <select> → dropdown
- <textarea> → large text

5. Output Description

Interactive form

6. Student Task

- Add more options

7. Real-World Project

Create **survey form**

LAB 13: Login Form

CODE

```
<!DOCTYPE html>
<html>
<body>

<h2>Login Form</h2>

<form>
  Username:<br>
  <input type="text"><br><br>

  Password:<br>
  <input type="password"><br><br>

  <input type="submit" value="Login">
</form>

</body>
</html>
```

Output:

Login Form

Username:

Password:

Login

Explanation

- password → hides text
- used for login system

Project

Simple school login page

LAB 14: Registration Form

CODE

```
<!DOCTYPE html>
<html>
<body>

<h2>Registration Form</h2>

<form>
  Name: <input type="text"><br><br>
  Email: <input type="email"><br><br>
  Password: <input type="password"><br><br>

  Gender:
  <input type="radio" name="g">Male
  <input type="radio" name="g">Female<br><br>

  <input type="submit" value="Register">
</form>

</body>
</html>
```

Output :

Registration Form

Name:

Email:

Password:

Gender: ☐ Male ☐ Female

Project

Student **registration** system UI

LAB 15: CSS Introduction

1. Objective

- Learn styling

2. Theory

CSS is used to style HTML

3. Syntax

```
selector {  
  property: value;  
}
```

Code:

//HTML code

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Introduction</title>

  <!-- Linking external CSS file -->
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- Main Heading -->
  <h1>Welcome to CSS</h1>

  <!-- Paragraph -->
  <p>This is a simple example of CSS styling.</p>

  <!-- Button -->
  <button>Click Me</button>

</body>
</html>
```

//css code

```
/* Style for body */
body {
  background-color: lightgray; /* page background color */
  font-family: Arial; /* font style */
}

/* Style for heading */
h1 {
  color: blue; /* text color */
}
```

```

    text-align: center; /* center alignment */
}

/* Style for paragraph */
p {
    color: green;
    font-size: 18px;
}

/* Style for button */
button {
    background-color: red;
    color: white;
    padding: 10px;
    border: none;
}

```

Welcome to CSS

This is a simple example of CSS styling.

Click Me

LAB 16: Inline CSS

<p style="color:red;">Red Text</p>

CSS inside tag

<!DOCTYPE html>

<html>

<head>

<title>Inline CSS Example</title>

```
</head>

<body style="background-color: lightgray; font-family: Arial;">

  <!-- Main Heading with inline CSS -->

  <h1 style="color: blue; text-align: center;">

    Welcome to CSS

  </h1>

  <!-- Paragraph with inline CSS -->

  <p style="color: green; font-size: 18px;">

    This is a simple example of CSS styling using inline CSS.

  </p>

  <!-- Button with inline CSS -->

  <button style="background-color: red; color: white; padding: 10px; border:
none;">

    Click Me

  </button>

</body>

</html>
```

Welcome to CSS

This is a simple example of CSS styling.

Click Me

LAB 17: Internal CSS

```
<style>
p {
  color: blue;
}
</style>
```

CSS inside <head>

CODE

```
<!DOCTYPE html>
<html>
<head>
  <title>Internal CSS Example</title>

  <!-- Internal CSS starts here -->
  <style>

    /* Style for entire page */
    body {
      background-color: lightgray; /* page background */
      font-family: Arial; /* font style */
    }

    /* Style for heading */
    h1 {
      color: blue; /* text color */
      text-align: center; /* center alignment */
    }

    /* Style for paragraph */
    p {
      color: green;
```

```
    font-size: 18px;
}
```

```
/* Style for button */
button {
    background-color: red;
    color: white;
    padding: 10px;
    border: none;
}
```

```
</style>
<!-- Internal CSS ends here -->
```

```
</head>
```

```
<body>
```

```
<h1>Welcome to CSS</h1>
```

```
<p>This is a simple example of CSS using internal styling.</p>
```

```
<button>Click Me</button>
```

```
</body>
```

```
</html>
```

LAB 18: External CSS

HTML

```
<link rel="stylesheet" href="style.css">
```

CSS file

```
p {  
  color: green;  
}
```

Separate file

First example

LAB 19: Styling Forms with CSS

HTML CODE: part 1

```
<!DOCTYPE html>  
<html>  
<head>  
  <title>Styled Form</title>  
  
  <!-- Linking external CSS -->  
  <link rel="stylesheet" href="style.css">  
</head>  
  
<body>  
  
  <h2>Student Form</h2>  
  
  <!-- Form starts -->  
  <form>
```

```
<!-- Name field -->
<label>Name:</label>
<input type="text" placeholder="Enter your name">

<!-- Email field -->
<label>Email:</label>
<input type="email" placeholder="Enter your email">

<!-- Password field -->
<label>Password:</label>
<input type="password" placeholder="Enter password">

<!-- Submit button -->
<input type="submit" value="Submit">

</form>

</body>
</html>
```

CSS PART - 2

```
/* Style for entire page */
body {
    font-family: Arial;
    background-color: lightgray;
}

/* Heading style */
h2 {
    text-align: center;
}

/* Form container */
```

```
form {  
    width: 300px;  
    margin: auto; /* center form */  
    background-color: white;  
    padding: 20px;  
    border-radius: 10px; /* rounded corners */  
}
```

```
/* Label style */  
label {  
    display: block; /* moves label to new line */  
    margin-top: 10px;  
}
```

```
/* Input fields */  
input {  
    width: 100%; /* full width */  
    padding: 8px;  
    margin-top: 5px;  
}
```

```
/* Submit button */  
input[type="submit"] {  
    background-color: blue;  
    color: white;  
    border: none;  
    margin-top: 15px;  
    cursor: pointer;  
}
```

```
/* Hover effect */  
input[type="submit"]:hover {  
    background-color: darkblue;  
}
```

LAB 20: Login Form using CSS (Final Project)